

Triangle Number Sums

Time limit: 1.00 s

Memory limit: 512 MB

A triangle number is a positive integer of the form $1 + 2 + \dots + k$. The first triangle numbers are 1, 3, 6, 10 and 15.

Every positive integer can be represented as a sum of triangle numbers. For example, $42 = 21 + 21$ and $1337 = 1326 + 10 + 1$.

Given a positive integer n , determine the smallest number of triangle numbers that sum to n .

Input

The first line has an integer t : the number of tests.

After that, each line has a positive integer n .

Output

For each test, print the smallest number of triangle numbers.

Constraints

- $1 \leq t \leq 100$
- $1 \leq n \leq 10^{12}$

Example

Input	Output
5	1
1	2
2	1
3	2
42	3
1337	